Define $l = \sqrt{\frac{\hbar}{m\omega}}$ as the oscillator's length scale

$$\dim\left(\frac{\hbar}{m\omega}\right) = \frac{J \cdot s}{kg \cdot \frac{1}{s}} = \frac{\cancel{kg} \cdot \frac{m}{\cancel{s^2}} \cdot m \cdot \cancel{s}}{\cancel{kg}/\cancel{s}} = m^2$$

$$\Rightarrow \dim(l) = \dim\sqrt{\frac{\hbar}{m\omega}} = \underline{\text{meters}} \quad \text{ok.}$$

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i}{m\omega}\hat{p}\right)$$

$$\hat{a}^+ = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i}{m\omega}\hat{p}\right)$$

$$m\omega = \frac{\hbar}{l^2} \Rightarrow \frac{1}{m\omega} = \frac{l^2}{\hbar} \qquad \sqrt{m\omega\hbar} = \sqrt{\frac{\hbar^2}{l^2}} = \frac{\hbar}{l}$$

$$\hat{a} = \frac{1}{\sqrt{2}}\left(\frac{\hat{x}}{l} + i\frac{l}{\hbar}\hat{p}\right)$$

$$\hat{a} = \frac{1}{\sqrt{2}}\left(\frac{\hat{x}}{l} - i\frac{l}{\hbar}\hat{p}\right)$$

$$\mathrm{Dim}\left(\frac{\hbar}{l}\right) = \frac{J \cdot s}{m} = \frac{kg \cdot m/\cancel{s^2} \cdot \cancel{m} \cdot \cancel{s}}{\cancel{m}} = kg \cdot \frac{m}{s} = \mathrm{Dim}(\hat{p}).$$

So that finally
$$\begin{cases} \hat{a} = \frac{1}{\sqrt{2}}\left(\frac{\hat{x}}{l} + i\frac{\hat{p}}{\hbar/l}\right) \\ \hat{a}^+ = \frac{1}{\sqrt{2}}\left(\frac{\hat{x}}{l} - i\frac{\hat{p}}{\hbar/l}\right) \end{cases}$$

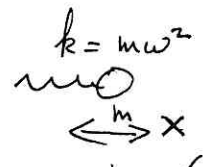
$$\begin{cases} \hat{x} = \frac{l}{\sqrt{2}}\left(\hat{a}^+ + \hat{a}\right) \\ \hat{p} = \frac{\hbar}{l}\frac{i}{\sqrt{2}}\left(\hat{a}^+ - \hat{a}\right) \end{cases}$$

Then $\psi_n(x) = \langle x|n\rangle = \langle x| \frac{(\hat{a}^+)^n}{\sqrt{n!}} |0\rangle =$

$$= \frac{1}{\sqrt{n!}} \langle x| \left[\frac{1}{\sqrt{2}}\left(\frac{\hat{x}}{\ell} - i\frac{\ell}{\hbar}\hat{p}\right)\right]^n |0\rangle = \left(\hat{p} = \frac{\hbar}{i}\frac{d}{dx}\right) =$$

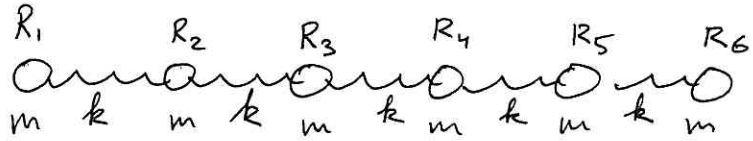
$$= \frac{1}{\sqrt{2^n n!}} \langle x| \left(\frac{\hat{x}}{\ell} - \ell\frac{d}{dx}\right)^n |0\rangle$$

Harmonic oscillator



$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 \longrightarrow \hbar\omega\left(a^+a + \frac{1}{2}\right)$$

Let us now consider the <u>chain of coupled oscillators</u>



$R_j = R_j^0 + x_j$

equilibrium — deviation

→ Classical energy $K.E. = \sum_{j=1}^{N} \frac{p_j^2}{2m}$

$$P.E. = \frac{1}{2}k\,(x_j - x_{j-1})^2 = \frac{1}{2}m\omega^2 \sum_{j=1}^{N}(x_j - x_{j-1})^2$$

$$\Rightarrow H = \sum_{j=1}^{N}\left(\frac{p_j^2}{2m} + \frac{1}{2}k\,(x_j - x_{j-1})^2\right)$$

only deviations from equilibrium <u>cost</u> energy. Also, uniform deviation does not cost anything.

→ the Lagrangian

$$L = KE - PE$$

$$L = \sum_{j=1}^{N}\left[\frac{p_j^2}{2m} - \frac{1}{2}k\,(x_j - x_{j-1})^2\right]$$

Lagrange equation of motion $(p = m\dot{x})$

$$\frac{\partial L}{\partial x_i} - \frac{d}{dt}\frac{\partial L}{\partial \dot{x}_i} = 0$$

$$\frac{\partial L}{\partial x_i} = -\frac{1}{2} k \frac{\partial}{\partial x_i} \sum_{j=1}^{N} (x_j - x_{j-1})^2 =$$

$$= -\frac{1}{2} k \frac{\partial}{\partial x_i} \left[(x_i - x_{i-1})^2 + (x_{i+1} - x_i)^2 \right] =$$

$$= -\frac{1}{2} k \frac{\partial}{\partial x_i} \left[x_i^2 - 2 x_i x_{i-1} + x_i^2 - 2 x_i x_{i+1} + \dots \right]$$

$$= -\frac{1}{2} k \left[2 x_i + 2 x_i - 2 x_{i-1} - 2 x_{i+1} \right] =$$

$$= -k \left(2 x_i - x_{i-1} - x_{i+1} \right)$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}_i} = \frac{d}{dt}\frac{\partial}{\partial \dot{x}_i}\left(\frac{m}{2}\dot{x}_j^2\right) = \frac{d}{dt} m \dot{x}_i = m \ddot{x}_i$$

$$\Rightarrow \quad m \ddot{x}_i = -k \left(2 x_i - x_{i-1} - x_{i+1} \right) = k \left(x_{i+1} + x_{i-1} - 2 x_i \right)$$

(classical equation of motion)

We now exploit periodicity $\quad x_j = x_{j+N}$

$$\cancel{x_j = \int_{-\pi}^{\pi} \frac{dq}{2\pi} e^{iqj} x_q}$$

(NOTE that realistically $N \sim 10^{23}$ atoms!)

~~The [illegible]~~

We use discrete Fourier transform $\left(a = R_j^0 - R_{j-1}^0 = \text{lattice spacing}\right)$

$$X_j = \frac{1}{\sqrt{N}} \sum_q X_q \, e^{iqja}$$

from $X_j = X_{j+N}$

$$\frac{1}{\sqrt{N}} \sum_q X_q \, e^{iqja} = \frac{1}{\sqrt{N}} \sum_q X_q \, e^{iq(j+N)a}$$

$$\Rightarrow \quad qNa = 2\pi n \quad , \; n = \text{Integer}$$

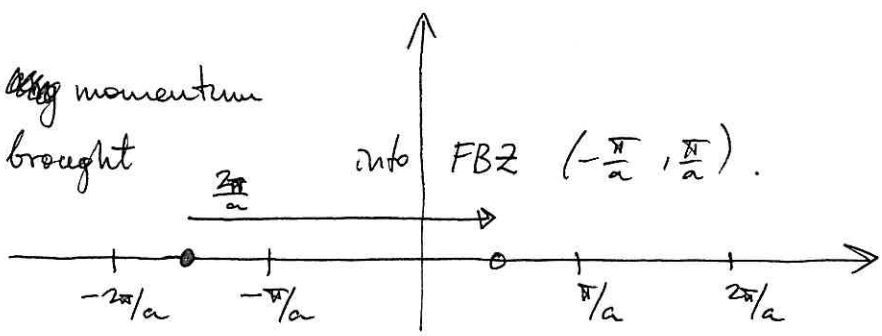
$$q = \frac{2\pi n}{Na} \qquad n = 0, \pm 1, \pm 2, \ldots \pm N$$

$$\Rightarrow \boxed{-\frac{\pi}{a} \le q \le \frac{\pi}{a}}$$ the First Brillouin Zone

This is because $e^{iqja} = e^{i(q + \frac{2\pi}{a})ja} = e^{iqja} e^{i(2\pi j)}$

but j = integer $\Rightarrow$ $\underline{e^{i2\pi j} = 1}$.

arbitrary momentum can be brought into FBZ $\left(-\frac{\pi}{a}, \frac{\pi}{a}\right)$.



Thus $$\frac{1}{\sqrt{N}} \sum_q m \ddot{X}_q \, e^{iqja} = \frac{1}{\sqrt{N}} \sum_q k \, X_q \left[e^{iq(j+1)a} + e^{iq(j-1)a} - 2 e^{iqja} \right] =$$

$$= \frac{1}{\sqrt{N}} \sum_q k \, X_q \, e^{iqja} \left(2 \cos qa - 2\right)$$

Seek stationary solution $X_q(t) = X_q\, e^{-iEt/\hbar}$

$$\dot{X} = -i\frac{E}{\hbar} X$$
$$\ddot{X} = -\frac{E^2}{\hbar^2} X$$

$\Rightarrow$ finally

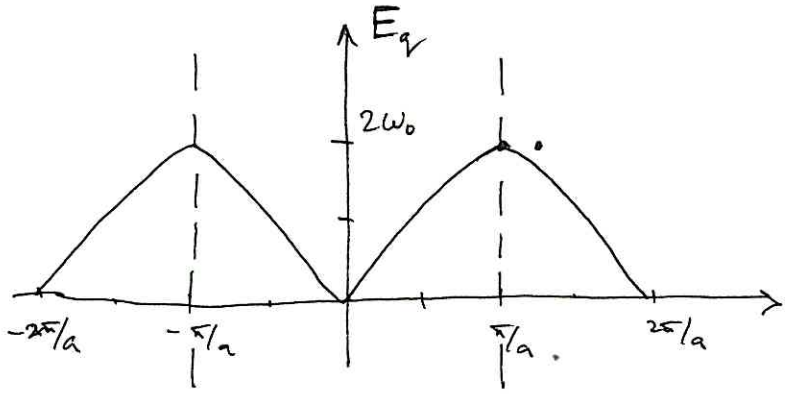
$$\frac{1}{\sqrt{N}} \sum_{q \in BZ} \left[-m\frac{E^2}{\hbar^2} + 2k(1-\cos qa) \right] X_q\, e^{iqja} = 0$$

$$\Rightarrow E_q^2 = \hbar^2 \frac{2k}{m}(1-\cos qa) = \hbar^2 \frac{4k}{m} \sin^2\frac{qa}{2}$$

$$\cos qa = \cos^2\frac{qa}{2} - \sin^2\frac{qa}{2} = 2\cos^2\frac{qa}{2} - 1 = 1 - 2\sin^2\frac{qa}{2}$$

$$E_q = \hbar 2\sqrt{\frac{k}{m}}\left|\sin\frac{qa}{2}\right| \qquad \left(\text{OR } k = m\omega_0^2\right)$$

$$E_q = \hbar 2\omega_0 \left|\sin\frac{qa}{2}\right| \Rightarrow \omega_q = 2\omega_0\left|\sin\frac{qa}{2}\right|.$$



Lattice oscillations = phonons:

For every momentum $q \in (-\pi/a, \pi/a)$ there is energy $E_q = 2\omega_0\left|\sin\frac{qa}{2}\right|$ that corresponds to propagating elastic wave of deformation.

For $q \to 0$ $\quad E_q \to 2\omega_0 \frac{qa}{2} = \omega_0 a q = \sqrt{\frac{k}{m}}\, a q$

describes propagating SOUND wave; $c = a\sqrt{\frac{k}{m}}$